

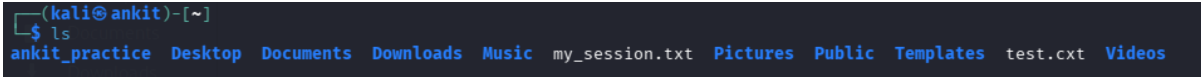
Name – Ankit Shringarpure

Intern Id- 2042

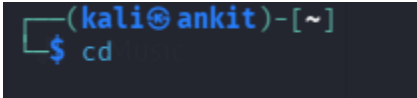
Linux commands

1) 

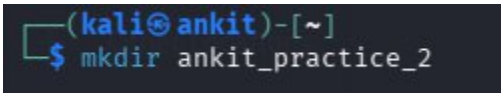
pwd - Print working directory

2) 

ls – List files and directories.

3) 

cd – change the current working directory in a command line interface.

4) 

mkdir – create a new directory.

5) 

cd Desktop – change the current directory to the desktop folder.

6)

```
(kali@ankit)-[~/Desktop]
$ touch test.cxt
```

Touch test.cxt – create an empty file or update the timestamp of an existing file.

7)

```
(kali@ankit)-[~/Desktop]
$ mv test.cxt final.txt
```

mv test.cxt final.txt – rename the file test.cxt to test.txt

8)

```
(kali@ankit)-[~/Desktop]
$ whoami
kali
```

Display the username of the currently logged-in user.

9)

```
(kali@ankit)-[~/Desktop]
$ uptime
05:58:40 up 2:02, 1 user, load average: 0.10, 0.17, 0.21
```

Uptime – shows how long the system has been running, along with the current load average.

10)

```
(kali@ankit)-[~/Desktop]
$ touch test.cxt
```

Touch test.cxt - create an empty file named test.cxt (or update its timestamp if it already exists).

11)

```
(kali@ankit)-[~/Desktop]
$ hostname
ankit
```

hostname – display the name of the of the current computer on the

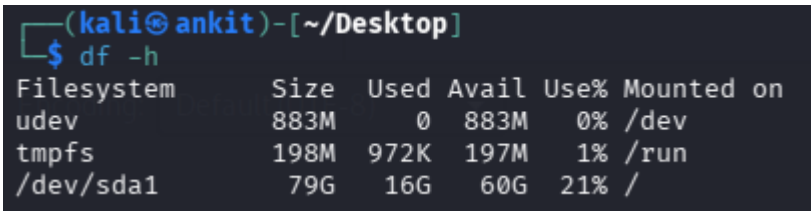
network

12) 

head test.cxt -display the first few lines of the file test.cxt (by default, the first 10 lines).

13) 

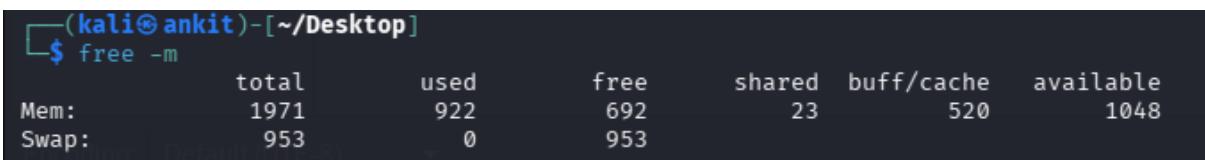
tail.test.cxt-display the last few lines of the file test.cxt (by default, the last 10 lines).

14) 

df -h - shows disk space usage of file systems in a human-readable format (KB, MB, GB).

15) 

Cat test.cxt - display the entire contents of the file test.cxt.

16) 

Free -m - displays the system's memory usage (RAM) in megabytes.

17)

```
(kali@ankit)-[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 71178sec preferred_lft 71178sec
    inet6 fd17:625c:f037:2:db7d:37ff:75bd:83ae/64 scope global dynamic noprefixroute
        valid_lft 86394sec preferred_lft 14394sec
    inet6 fe80::cfe8:d8bd:201a:a254/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Ip a - display all IP addresses and network interfaces on the system.

```
(kali@ankit)-[~/Desktop]
$ ping google.com
PING google.com (142.250.70.110) 56(84) bytes of data:
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=1 ttl=255 time=8.49 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=2 ttl=255 time=11.2 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=3 ttl=255 time=9.16 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=4 ttl=255 time=8.15 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=5 ttl=255 time=8.58 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=6 ttl=255 time=27.8 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=7 ttl=255 time=10.2 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=8 ttl=255 time=8.93 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7010ms
rtt min/avg/max/mdev = 8.146/11.563/27.802/6.208 ms
```

Ping google.com - check network connectivity to Google by sending test packets and receiving replies.

```
(kali@ankit)-[~/Desktop]
$ date
Wed Dec 31 10:09:03 AM EST 2025
```

19)

date – display the current system date and time.

20)

```
(kali㉿ ankit)-[~/Desktop]
$ uptime
10:20:52 up 6:24, 1 user, load average: 0.36, 0.23, 0.14
```

uptime - shows how long the system has been running and the current load average.

21)

```
(kali㉿ ankit)-[~]
$ whois
Usage: whois [OPTION]... OBJECT...

-h HOST, --host HOST    connect to server HOST
-p PORT, --port PORT    connect to PORT
-I                      query whois.iana.org and follow its referral
-H                      hide legal disclaimers
    --verbose           explain what is being done
    --no-recursion      disable recursion from registry to registrar servers
    --help              display this help and exit
    --version           output version information and exit

These flags are supported by whois.ripe.net and some RIPE-like servers:
-l                  find the one level less specific match
-L                  find all levels less specific matches
-m                  find all one level more specific matches
-M                  find all levels of more specific matches
-c                  find the smallest match containing a mnt-irt attribute
-x                  exact match
-b                  return brief IP address ranges with abuse contact
-B                  turn off object filtering (show email addresses)
-G                  turn off grouping of associated objects
-d                  return DNS reverse delegation objects too
-i ATTR[,ATTR] ...  do an inverse look-up for specified ATTRibutes
-T TYPE[,TYPE] ...  only look for objects of TYPE
-K                  only primary keys are returned
-r                  turn off recursive look-ups for contact information
-R                  force to show local copy of the domain object even
                    if it contains referral
-a                  also search all the mirrored databases
-s SOURCE[,SOURCE] ... search the database mirrored from SOURCE
-g SOURCE:FIRST-LAST find updates from SOURCE from serial FIRST to LAST
-t TYPE             request template for object of TYPE
-v TYPE             request verbose template for object of TYPE
-q [version|sources|types] query specified server info
```

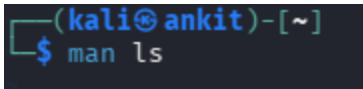
Whois- used to get registration and ownership information about a domain name or IP address.

22)

```
(kali@ankit)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:db7d:37ff:75bd:83ae prefixlen 64 scopeid 0<global>
    inet6 fe80::cff8:d8bd:201a:a254 prefixlen 64 scopeid 0<link>
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)
    RX packets 339 bytes 36587 (35.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 433 bytes 43833 (42.8 KiB)
    TX errors 0 dropped 15 overruns 0 carrier 0 collisions 0

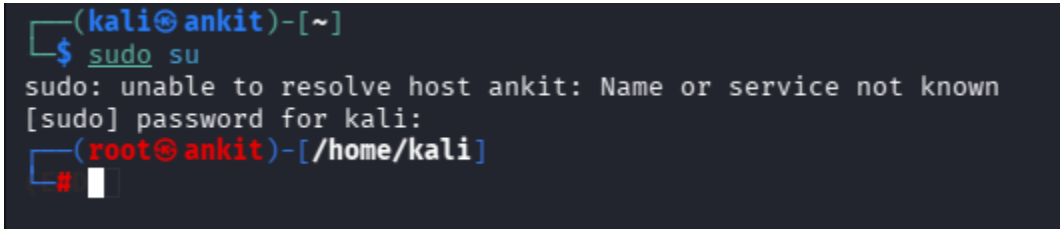
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
```

Ifconfig - display and configure network interface settings.

23) 

```
ls(1) User Commands
NAME
  ls - list directory contents
SYNOPSIS
  ls [OPTION]... [FILE]...
DESCRIPTION
  List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.
  Mandatory arguments to long options are mandatory for short options too.
  -a, --all
    do not ignore entries starting with .
  -A, --almost-all
    do not list implied . and ..
  --author
    with -l, print the author of each file
  -b, --escape
    print C-style escapes for nongraphical characters
  --block-size=SIZE
    with -l, scale sizes by SIZE when printing them; e.g., '--block size=M'; see SIZE format below
  -B, --ignore-backups
    do not list implied entries ending with ~
  -c
    with -lt: sort by, and show, time (time of last change of file status information); with -l: show time and sort by name; otherwise: sort by time, newest first
  -C
    list entries by columns
  --color[=WHEN]
    color the output WHEN; more info below
  -d, --directory
    list directories themselves, not their contents
  -D, --dired
    generate output designed for Emacs' dired mode
  -f
    sort as -a -u
  -F, --classify[=WHEN]
    append indicator (one of */=>@) to entries WHEN
  --file-type
    likewise, except do not append '*'
Manual page ls(1) line 1 (press h for help or q to quit)
```

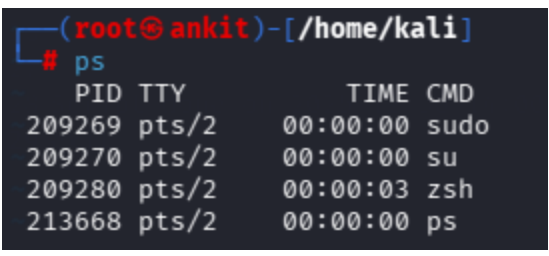
man ls - display the manual (help documentation) for the ls command.

24) 

sudo su- switch to the superuser (root) account with administrative privileges.

25) A terminal window with a dark background. The prompt is `(root@ankit)-[/home/kali]`. The user has entered `# echo "hello everyone"` and the output is `hello everyone`.

echo - display a line of text or the value of variables in the terminal.

26) A terminal window with a dark background. The prompt is `(root@ankit)-[/home/kali]`. The user has entered `# ps`. The output is a table of active processes:

PID	TTY	TIME	CMD
209269	pts/2	00:00:00	sudo
209270	pts/2	00:00:00	su
209280	pts/2	00:00:03	zsh
213668	pts/2	00:00:00	ps

ps - display information about active (running) processes on the system.